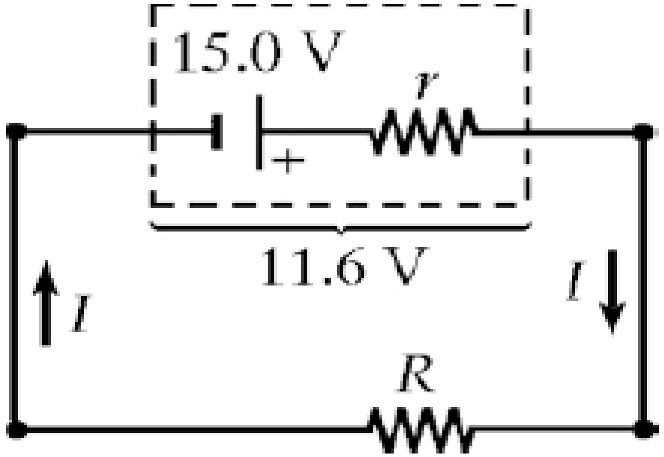
Solutions of Problems on CH 28

1. **M** A battery has an emf of 15.0 V. The terminal voltage of the battery is 11.6 V when it is delivering 20.0 W of power to an external load resistor R. (a) What is the value of R? (b) What is the internal resistance of the battery?

(a) $R = \dfrac{(\Delta V)^2}{P} = \dfrac{(11.6\text{ V})^2}{20.0\text{ W}} = \boxed{6.73\ \Omega}$



(b) $\mathcal{E} = IR + Ir$, where $I = \Delta V / R$.

$$r = \frac{(\mathcal{E} - IR)}{I} = \frac{(\mathcal{E} - \Delta V)R}{\Delta V}$$

$$= \frac{(15.0\text{ V} - 11.6\text{ V})(6.73\ \Omega)}{11.6\text{ V}} = \boxed{1.97\ \Omega}$$

5. **W** Three 100-Ω resistors are connected as shown in Figure P28.5. The maximum power that can safely be delivered to any one resistor is 25.0 W. (a) What is the maximum potential difference that can be applied to the terminals *a* and *b*? (b) For the voltage determined in part (a), what is the power delivered to each resistor? (c) What is the total power delivered to the combination of resistors?

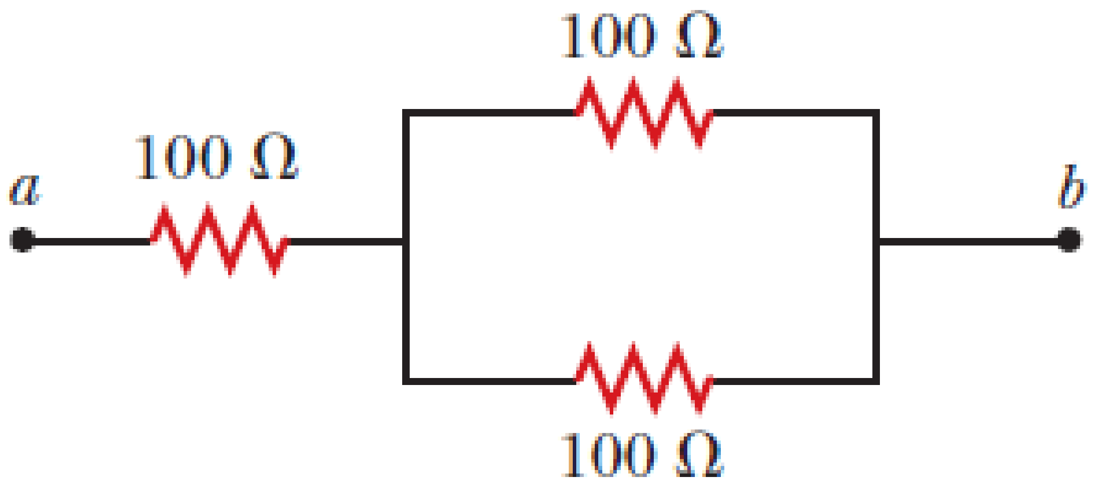
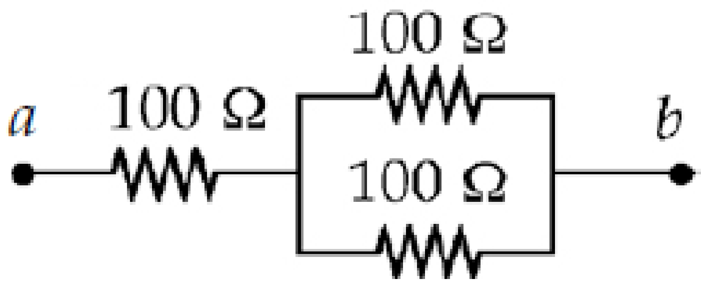


Figure P28.5

(a) Since all the current in the circuit must pass through the series 100-Ω resistor,



ANS. FIG. P28.5

$$P_{\max} = I_{\max}^2 R$$

so $$I_{\max} = \sqrt{\frac{P}{R}} = \sqrt{\frac{25.0\ \text{W}}{100\ \Omega}} = 0.500\ \text{A}.$$

$$R_{eq} = 100\ \Omega + \left(\frac{1}{100} + \frac{1}{100}\right)^{-1}\Omega = 150\ \Omega$$

$$\Delta V_{\max} = R_{eq} I_{\max} = \boxed{75.0\ \text{V}}$$

(b) From a to b in the circuit, the power delivered is

$$P_{\text{series}} = \boxed{25.0\ \text{W}}$$ for the first resistor, and

$$P_{\text{parallel}} = I^2R = (0.250\ \text{A})^2(100\ \Omega) = \boxed{6.25\ \text{W}}$$

for each of the two parallel resistors.

(c) $P = I\Delta V = (0.500\ \text{A})(75.0\ \text{V}) = \boxed{37.5\ \text{W}}$

9. **M** Consider the circuit shown in Figure P28.9. Find (a) the current in the 20.0-Ω resistor and (b) the potential difference between points *a* and *b*.

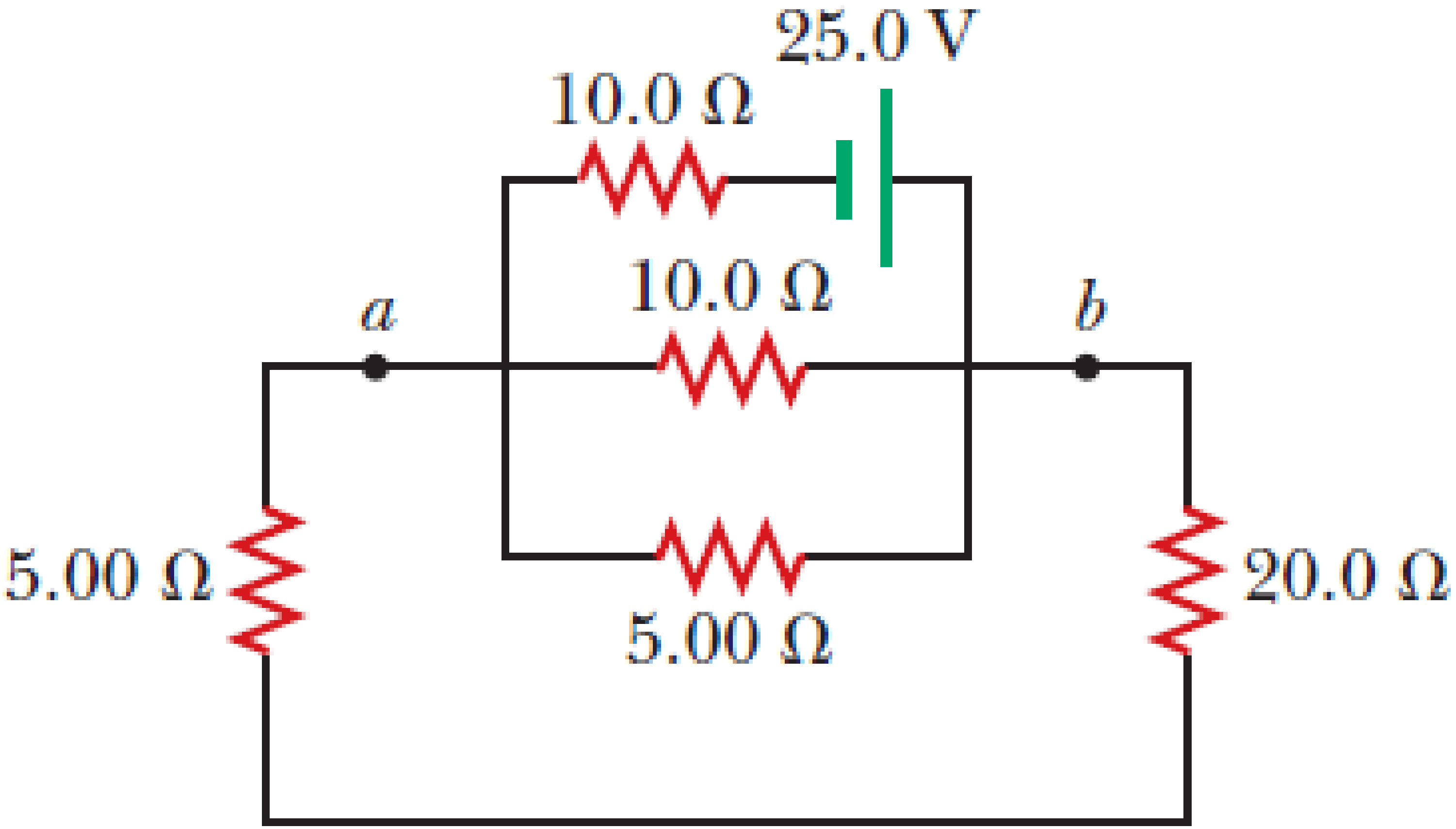
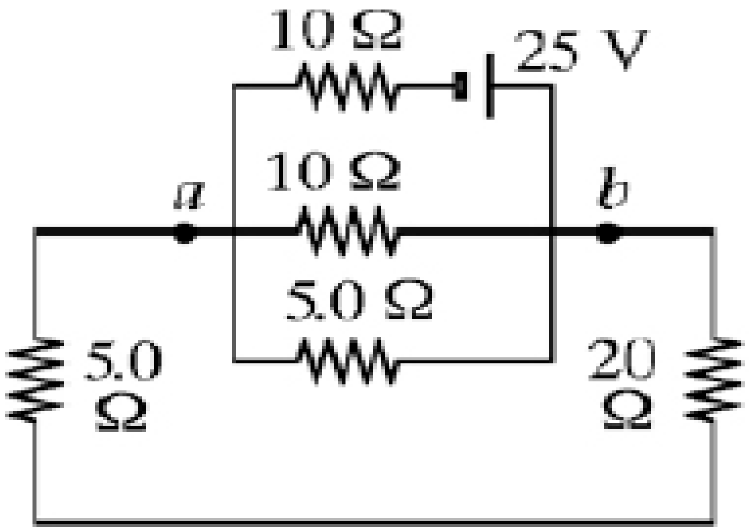
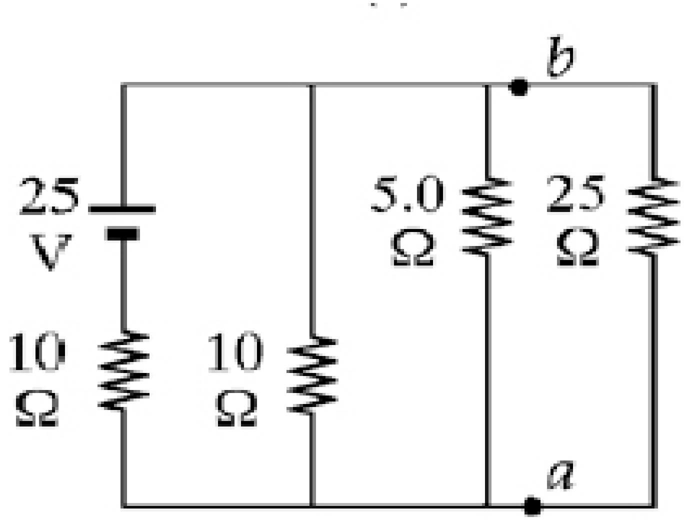


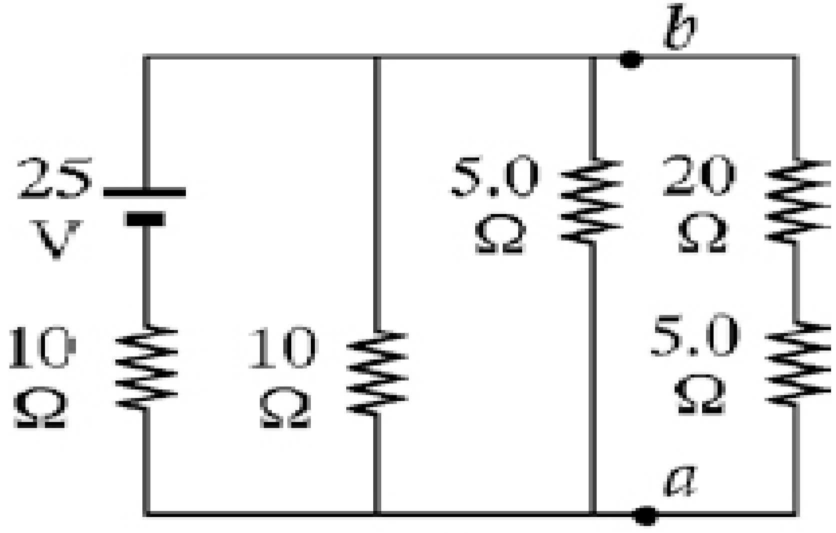
Figure P28.9

$$\frac{1}{R_{eq}} = \frac{1}{10.0\ \Omega} + \frac{1}{5.00\ \Omega} + \frac{1}{25.0\ \Omega} \rightarrow R_{eq} = 2.94\ \Omega$$

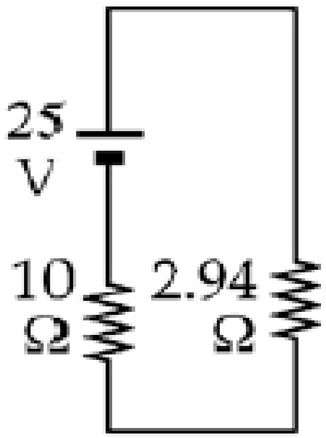




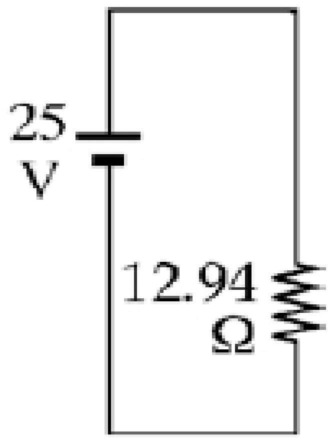
(b)



(a)



(c)



(d)

$$I = \frac{\Delta V}{R} = \frac{25.0\ \text{V}}{12.94\ \Omega} = 1.93\ \text{A}$$

$$\Delta V = IR = (1.93\ \text{A})(2.94\ \Omega) = 5.68\ \text{V}$$

$$\Delta V_{ab} = \boxed{5.68\ \text{V}}$$

$$I = \frac{\Delta V_{ab}}{R_{ab}} = \frac{5.68\ \text{V}}{25.0\ \Omega} = 0.227\ \text{A} = \boxed{227\ \text{mA}}$$

13. M (a) Find the equivalent resistance between points a and b in Figure P28.13. (b) Calculate the current in each resistor if a potential difference of 34.0 V is applied between points a and b.

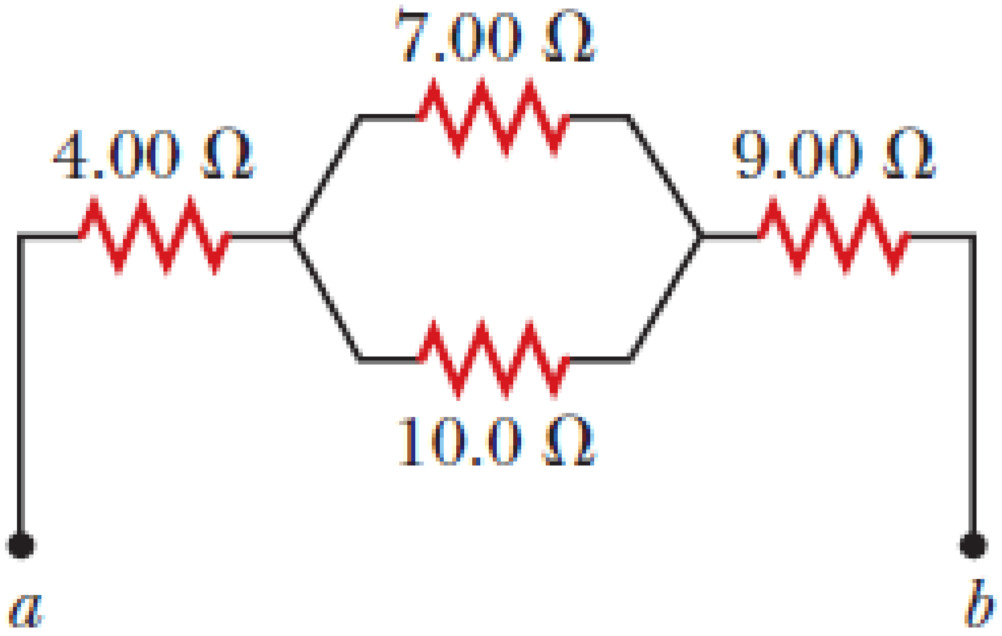


Figure P28.13

(a) $$R_p = \frac{1}{(1/7.00\ \Omega)+(1/10.0\ \Omega)} = 4.12\ \Omega$$

$$R_s = R_1 + R_2 + R_3 = 4.00 + 4.12 + 9.00 = \boxed{17.1\ \Omega}$$

(b) $$\Delta V = IR$$

$$34.0\ \text{V} = I(17.1\ \Omega)$$

$I = \boxed{1.99\ \text{A}}$ for the $4.00\text{-}\Omega$ and $9.00\text{-}\Omega$ resistors.

Applying $\Delta V = IR$, $(1.99\ \text{A})(4.12\ \Omega) = 8.18\ \text{V}$

$$8.18\ \text{V} = I(7.00\ \Omega)$$

$I = \boxed{1.17\ \text{A}}$ for the $7.00\text{-}\Omega$ resistor.

$$8.18\ \text{V} = I(10.0\ \Omega)$$

$I = \boxed{0.818\ \text{A}}$ for the $10.0\text{-}\Omega$ resistor.

15. Two resistors connected in series have an equivalent resistance of 690 Ω. When they are connected in parallel, their equivalent resistance is 150 Ω. Find the resistance of each resistor.

$$x + y = 690, \text{ and } \frac{1}{150} = \frac{1}{x} + \frac{1}{y}$$

$$\frac{1}{150} = \frac{1}{x} + \frac{1}{690 - x} = \frac{(690 - x) + x}{x(690 - x)}$$

$$x^2 - 690x + 103\,500 = 0$$

$$x = \frac{690 \pm \sqrt{(690)^2 - 414\,000}}{2}$$

$$x = \boxed{470\ \Omega} \qquad y = \boxed{220\ \Omega}$$

17. Consider the combination of resistors shown in Figure P28.17. (a) Find the equivalent resistance between points *a* and *b*. (b) If a voltage of 35.0 V is applied between points *a* and *b*, find the current in each resistor.

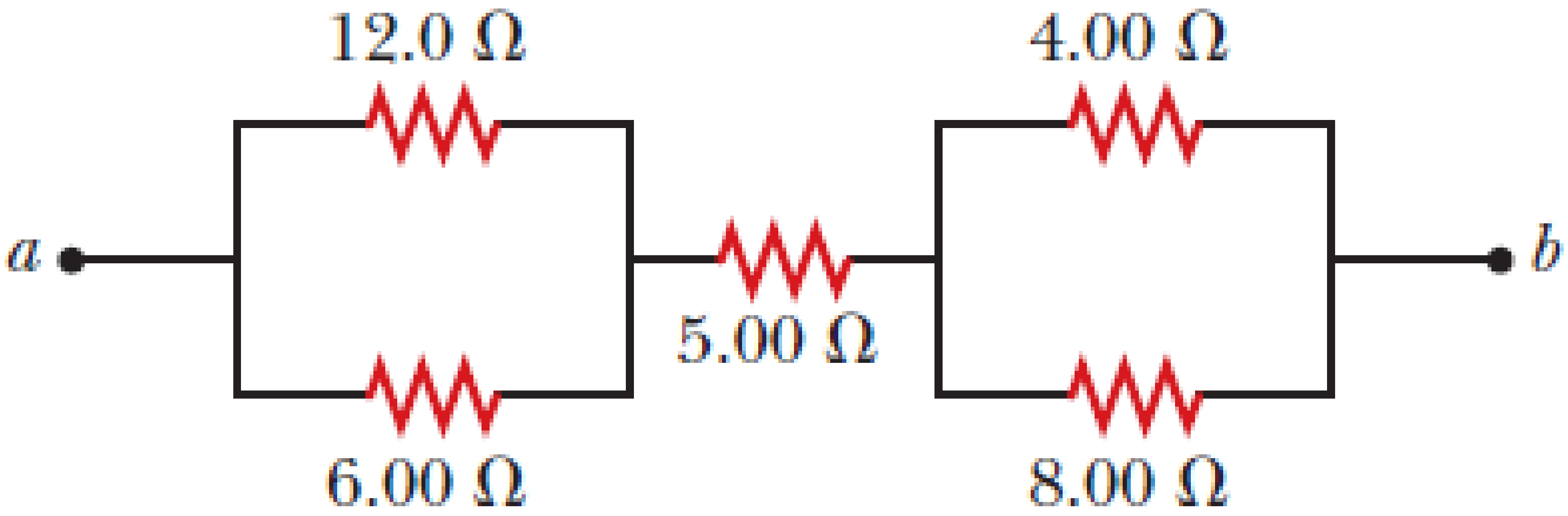


Figure P28.17

(a) The parallel combination of the 6.0 Ω and 12 Ω resistors has an equivalent resistance of

$$\frac{1}{R_{p1}} = \frac{1}{6.0\ \Omega} + \frac{1}{12\ \Omega} = \frac{2+1}{12\ \Omega}$$

or $$R_{p1} = \frac{12\ \Omega}{3} = 4.0\ \Omega$$

Similarly, the equivalent resistance of the 4.0 Ω and 8.0 Ω parallel combination is

$$\frac{1}{R_{p2}} = \frac{1}{4.0\ \Omega} + \frac{1}{8.0\ \Omega} = \frac{2+1}{8.0\ \Omega}$$

or $$R_{p2} = \frac{8\ \Omega}{3}$$

The total resistance of the series combination between points a and b is then

$$R_{ab} = R_{p1} + 5.0\ \Omega + R_{p2} = 4.0\ \Omega + 5.0\ \Omega + \frac{8.0}{3}\ \Omega$$

$$= \frac{35}{3}\ \Omega = \boxed{11.7\ \Omega}$$

(b) If $\Delta V_{ab} = 35$ V, the total current from a to b is $I_{ab} = \Delta V_{ab}/R_{ab} =$ $35\text{ V}/(35\ \Omega/3) = 3.0\text{ A}$ and the potential differences across the two parallel combinations are

$$\Delta V_{p1} = I_{ab}R_{p1} = (3.0\text{ A})(4.0\ \Omega) = 12\text{ V}$$

and $$\Delta V_{p2} = I_{ab}R_{p2} = (3.0\text{ A})\left(\frac{8.0}{3}\ \Omega\right) = 8.0\text{ V}$$

so the individual currents through the various resistors are:

$$I_{12} = \Delta V_{p1}/12\ \Omega = \boxed{1.0\text{ A}}$$

$$I_6 = \Delta V_{p1}/6.0\ \Omega = \boxed{2.0\text{ A}}$$

$$I_5 = I_{ab} = \boxed{3.0\text{ A}}$$

$$I_8 = \Delta V_{p2}/8.0\ \Omega = \boxed{1.0\text{ A}}$$

and $$I_4 = \Delta V_{p2}/4.0\ \Omega = \boxed{2.0\text{ A}}$$

19. **W** Calculate the power delivered to each resistor in the circuit shown in Figure P28.19.

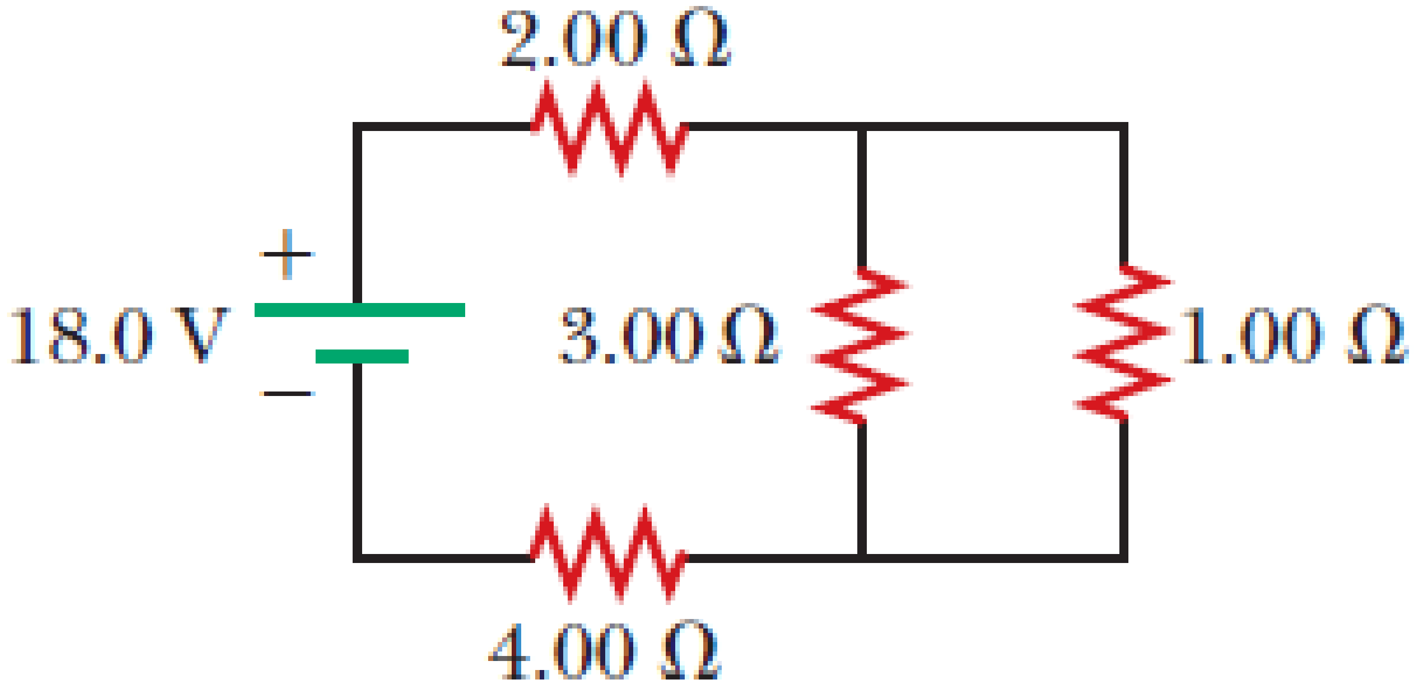


Figure P28.19

To find the current in each resistor, we find the resistance seen by the battery. The given circuit reduces as shown in ANS. FIG. P28.19(a), since

$$\frac{1}{(1/1.00\ \Omega)+(1/3.00\ \Omega\ )}=0.750\ \Omega$$

In ANS. FIG. P28.19(b),

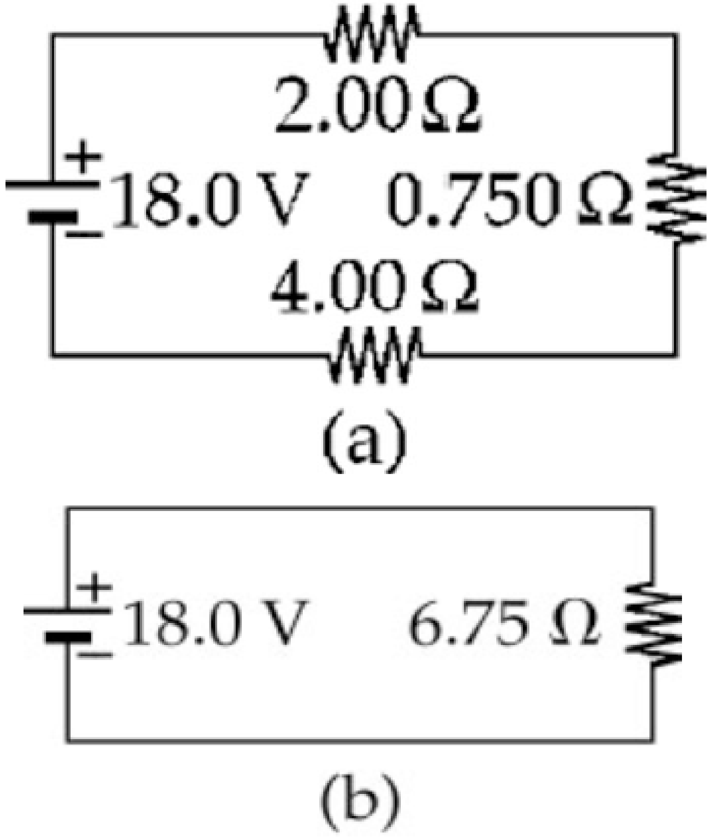
$$I=18.0\ \mathrm{V}/6.75\ \Omega=2.67\ \mathrm{A}$$

This is also the current in ANS. FIG. P28.19(a), so the 2.00-Ω and 4.00-Ω resistors convert powers

$$P_2=I\Delta V=I^2R=(2.67\ \mathrm{A})^2(2.00\ \Omega)=\boxed{14.2\ \mathrm{W}}$$

and $P_4=I^2R=(2.67\ \mathrm{A})^2(4.00\ \Omega)=\boxed{28.4\ \mathrm{W}}$

The voltage across the 0.750-Ω resistor in ANS. FIG. P28.19(a), and across both the 3.00-Ω and the 1.00-Ω resistors in Figure P28.19, is



ANS. FIG. P28.19

$$\Delta V = IR = (2.67\ \text{A})(0.750\ \Omega) = \boxed{2.00\ \text{V}}$$

Then for the 3.00-Ω resistor,

$$I = \frac{\Delta V}{R} = \frac{2.00\ \text{V}}{3.00\ \Omega}$$

and the power is

$$P_3 = I\Delta V = \left(\frac{2.00\ \text{V}}{3.00\ \Omega}\right)(2.00\ \text{V}) = \boxed{1.33\ \text{W}}$$

For the 1.00-Ω resistor,

$$I = \frac{2.00\ \text{V}}{1.00\ \Omega} \quad \text{and} \quad P_1 = \left(\frac{2.00\ \text{V}}{1.00\ \Omega}\right)(2.00\ \text{V}) = \boxed{4.00\ \text{W}}$$

23. The circuit shown in Figure P28.22 is connected for 2.00 min. (a) Determine the current in each branch of the circuit. (b) Find the energy delivered by each battery. (c) Find the energy delivered to each resistor. (d) Identify the type of energy storage transformation that occurs in the operation of the circuit. (e) Find the total amount of energy transformed into internal energy in the resistors.

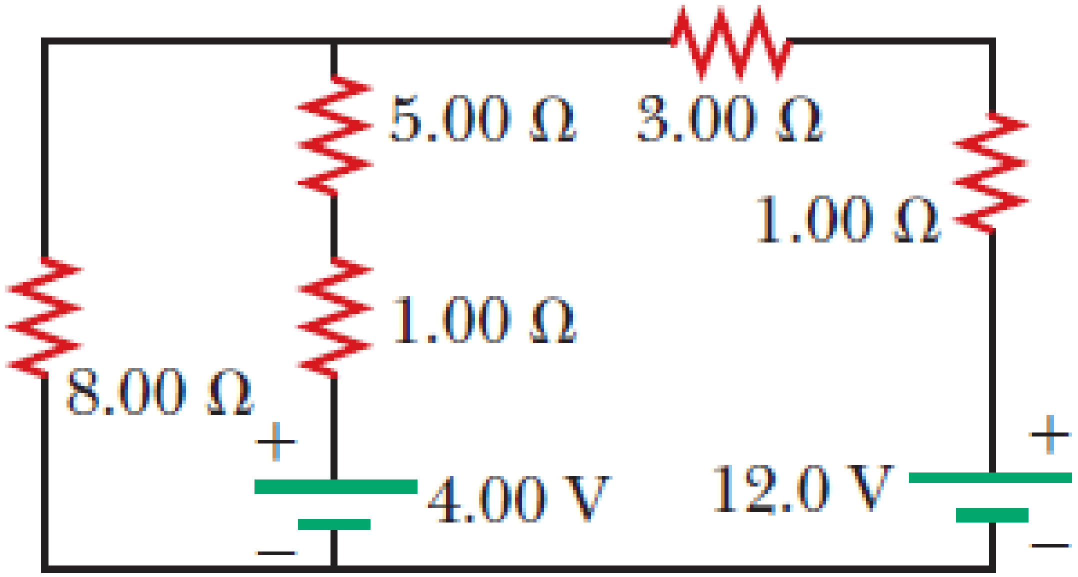
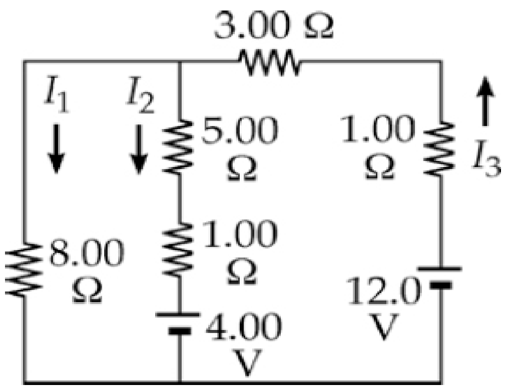


Figure P28.22 Problems 22 and 23.

We name currents I_1, I_2, and I_3 as shown in ANS. FIG. P28.23. From Kirchhoff's current rule, $I_3 = I_1 + I_2$.



ANS. FIG. P28.23

Applying Kirchhoff's voltage rule to the loop containing I_2 and I_3,

$$12.0\text{ V} - (4.00\ \Omega)I_3 - (6.00\ \Omega)I_2 - 4.00\text{ V} = 0$$

$$8.00 = (4.00)I_3 + (6.00)I_2$$

Applying Kirchhoff's voltage rule to the loop containing I_1 and I_2,

$$-(6.00\ \Omega)I_2 - 4.00\text{ V} + (8.00\ \Omega)I_1 = 0$$

or $$(8.00\ \Omega)I_1 = 4.00 + (6.00\ \Omega)I_2$$

Solving the above linear system (by substituting $I_1 + I_2$ for I_3), we proceed to the pair of simultaneous equations:

$$\begin{cases} 8 = 4I_1 + 4I_2 + 6I_2 \\ 8I_1 = 4 + 6I_2 \end{cases} \quad \text{or} \quad \begin{cases} 8 = 4I_1 + 10I_2 \\ I_2 = \dfrac{4}{3}I_1 - \dfrac{2}{3} \end{cases}$$

and to the single equation

$$8 = 4I_1 + 10\left(\frac{4}{3}I_1 - \frac{2}{3}\right) = \frac{52}{3}I_1 - \frac{20}{3}$$

which gives

$$I_1 = \frac{3}{52}\left(8 + \frac{20}{3}\right) = 0.846 \text{ A}$$

Then $$I_2 = I_2 = \frac{4}{3}(0.846) - \frac{2}{3} = 0.462$$

and $$I_3 = I_1 + I_2 = 1.31 \text{ A}$$

give $$\boxed{I_1 = 846 \text{ mA}, I_2 = 462 \text{ mA}, I_3 = 1.31 \text{ A}}$$

(a) The results are: $\boxed{\text{0.846 A down in the 8.00-}\Omega\text{ resistor; 0.462 A down in the middle branch; 1.31 A up in the right-hand branch}}$.

(b) For 4.00-V battery:

$$\Delta U = P\Delta t = (\Delta V)I\Delta t = (4.00\ \text{V})(-0.462\ \text{A})(120\ \text{s}) = -222\ \text{J}$$

For 12.0-V battery:

$$\Delta U = (12.0\ \text{V})(1.31\ \text{A})(120\ \text{s}) = 1.88\ \text{kJ}$$

The results are: $\boxed{-222\text{ J by the 4.00-V battery and 1.88 kJ by the 12.0-V battery}}$.

(c) To the 8.00-Ω resistor:

$$\Delta U = I^2 R \Delta t = (0.846\ \text{A})^2 (8.00\ \Omega)(120\ \text{s}) = \boxed{687\ \text{J}}$$

To the 5.00-Ω resistor:

$$\Delta U = (0.462\ \text{A})^2 (5.00\ \Omega)(120\ \text{s}) = \boxed{128\ \text{J}}$$

To the 1.00-Ω resistor in the center branch:

$$(0.462\ \text{A})^2 (1.00\ \Omega)(120\ \text{s}) = \boxed{25.6\ \text{J}}$$

To the 3.00-Ω resistor:

$$(1.31\ \text{A})^2 (3.00\ \Omega)(120\ \text{s}) = \boxed{616\ \text{J}}$$

To the 1.00-Ω resistor in the right-hand branch:

$$(1.31\ \text{A})^2 (1.00\ \Omega)(120\ \text{s}) = \boxed{205\ \text{J}}$$

(d) Chemical energy in the 12.0-V battery is transformed into internal energy in the resistors. The 4.00-V battery is being charged, so its chemical potential energy is increasing at the expense of some of the chemical potential energy in the 12.0-V battery.

(e) Either sum the results in part (b): $-222\ \text{J} + 1.88\ \text{kJ} = 1.66\ \text{kJ}$,

or in part (c): $687\ \text{J} + 128\ \text{J} + 25.6\ \text{J} + 616\ \text{J} + 205\ \text{J} = 1.66\ \text{kJ}$

The total amount of energy transformed is $\boxed{1.66\ \text{kJ}}$.

25. **M** What are the expected readings of (a) the ideal ammeter and (b) the ideal voltmeter in Figure P28.25?

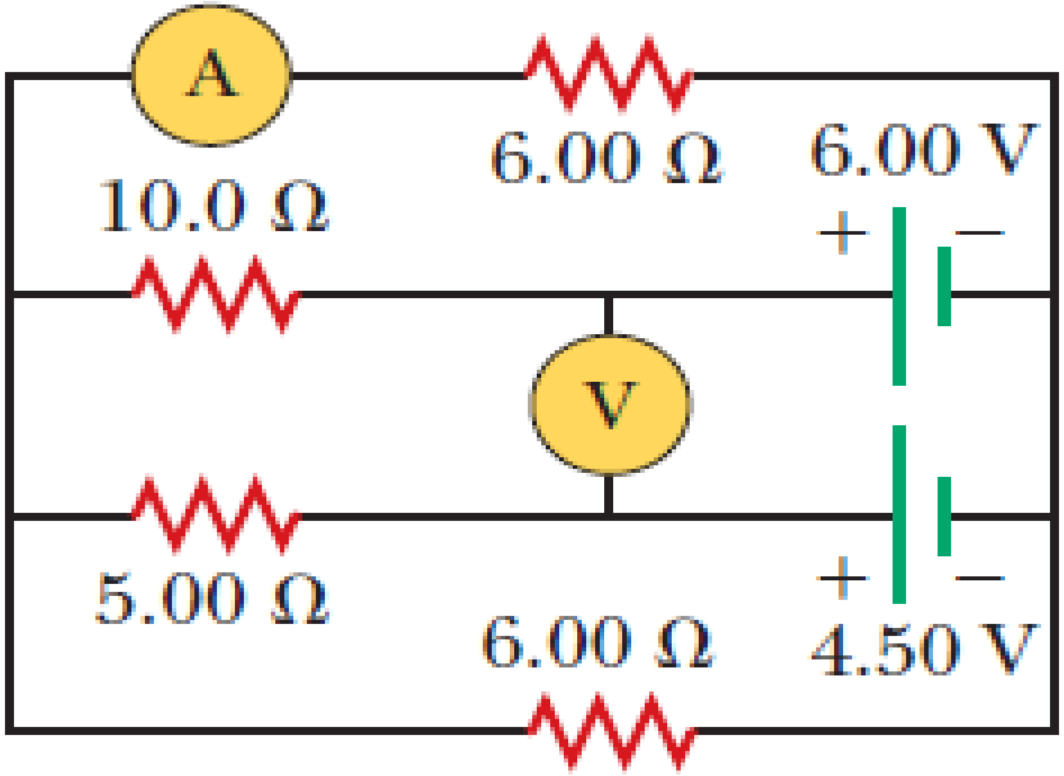


Figure P28.25

(a) Let I_6 represent the current in the ammeter and the top 6-Ω resistor. The bottom 6-Ω resistor has the same potential difference across it, so it carries an equal current.

We assume both I_6 in the upper branch and I_6 in the lower branch flow to the right. We assume current I_{10} flows to the left through the 10-Ω resistor. For the top loop we have

$$6.00 - 10.0I_{10} - 6.00I_6 = 0 \quad \rightarrow \quad I_{10} = 0.6 - 0.6\, I_6 \qquad \textbf{[1]}$$

We assume current I_5 flows to the left through the 5-Ω resistor. For the bottom loop,

$$4.50 - 5.00I_5 - 6.00I_6 = 0 \quad \rightarrow \quad I_5 = 0.9 - 1.2\, I_6 \qquad \textbf{[2]}$$

For the junctions on the left side, taken together,

$$+I_{10} + I_5 - I_6 - I_6 = 0 \qquad \textbf{[3]}$$

Substituting I_{10} and I_5 into [3], we have

$$(0.6 - 0.6\, I_6) + (0.9 - 1.2\, I_6) - 2\, I_6 = 0 \rightarrow I_6 = 1.5/3.8 = \boxed{0.395 \text{ A}}$$

(b) The loop theorem for the little loop containing the voltmeter gives

$$+\, 6.00 \text{ V} - \Delta V - 4.50 \text{ V} = 0 \quad \rightarrow \quad \Delta V = \boxed{1.50\ V}$$

29. The ammeter shown in Figure P28.29 reads 2.00 A. **W** Find (a) I_1, (b) I_2, and (c) $\mathcal{E}$.

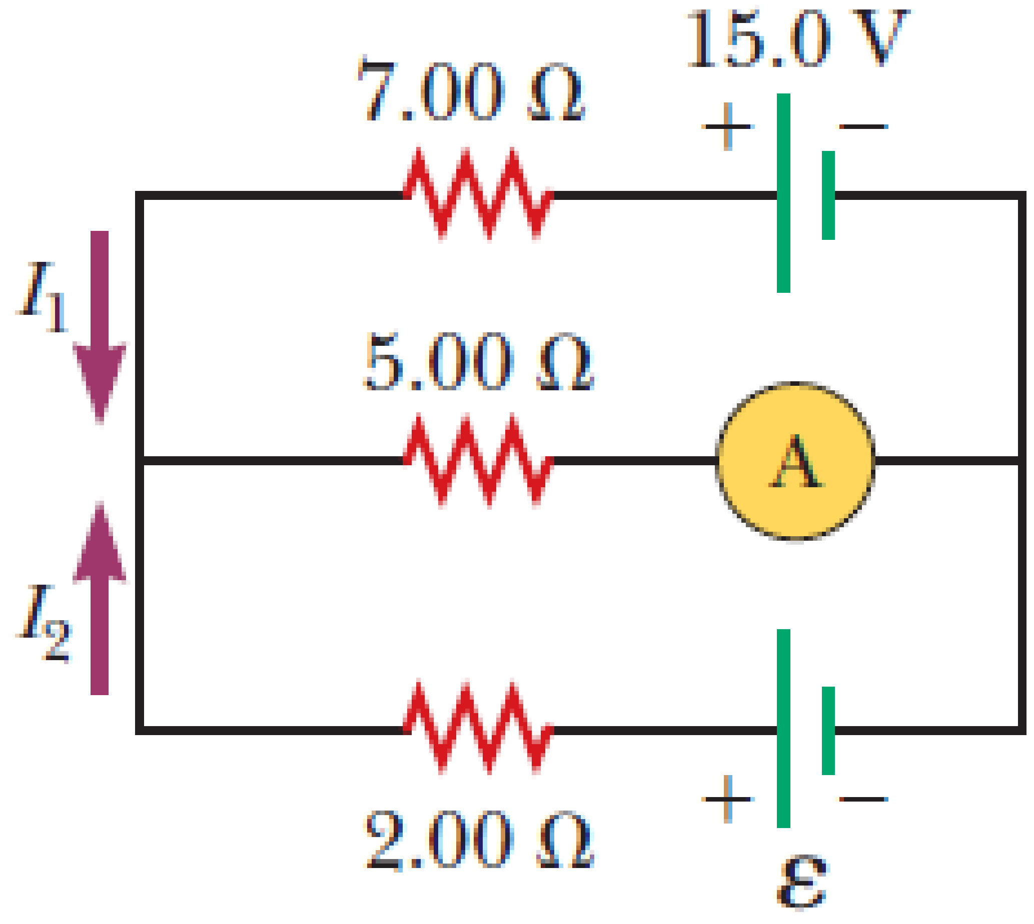
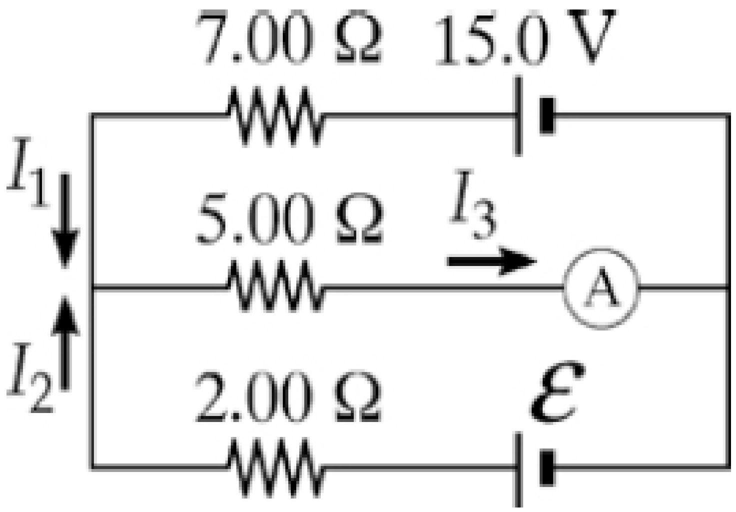


Figure P28.29

(a) For the upper loop:

$$+15.0\text{ V} - (7.00\ \Omega)I_1 - (2.00\text{ A})(5.00\ \Omega) = 0$$

$$5.00 = 7.00 I_1 \quad \text{so} \quad \boxed{I_1 = 0.714\text{ A}}$$



(b) For the center-left junction:

$$I_3 = I_1 + I_2 = 2.00\text{ A}$$

$$0.714 + I_2 = 2.00 \quad \text{so} \quad \boxed{I_2 = 1.29\text{ A}}$$

(c) For the lower loop:

$$+\mathcal{E} - (2.00\ \Omega)(1.29\text{ A}) - (5.00\ \Omega)(2.00\text{ A}) = 0 \rightarrow \boxed{\mathcal{E} = 12.6\text{ V}}$$

37. An uncharged capacitor and a resistor are connected in series to a source of emf. If $\mathcal{E} = 9.00$ V, $C = 20.0\ \mu$F, and $R = 100\ \Omega$, find (a) the time constant of the circuit, (b) the maximum charge on the capacitor, and (c) the charge on the capacitor at a time equal to one time constant after the battery is connected.

(a) The time constant of the circuit is

$$\tau = RC = (100\ \Omega)(20.0\times10^{-6}\ \text{F}) = 2.00\times10^{-3}\ \text{s} = \boxed{2.00\ \text{ms}}$$

(b) The maximum charge on the capacitor is given by Equation 28.13:

$$Q_{\max} = C\mathcal{E} = (20.0\times10^{-6}\ \text{F})(9.00\ \text{V}) = \boxed{1.80\times10^{-4}\ \text{C}}$$

(c) We use $q(t) = Q_{\max}\left(1 - e^{-t/RC}\right)$, when $t = RC$. Then,

$$q(t) = Q_{\max}\left(1 - e^{-RC/RC}\right) = Q_{\max}\left(1 - e^{-1}\right) = (1.80\times10^{-4}\ \text{C})\left(1 - e^{-1}\right)$$

$$= \boxed{1.14\times10^{-4}\ \text{C}}$$

38. Consider a series RC circuit as in Figure P28.38 for which $R = 1.00\ \text{M}\Omega$, $C = 5.00\ \mu\text{F}$, and $\mathcal{E} = 30.0$ V. Find (a) the time constant of the circuit and (b) the maximum charge on the capacitor after the switch is thrown closed. (c) Find the current in the resistor 10.0 s after the switch is closed.

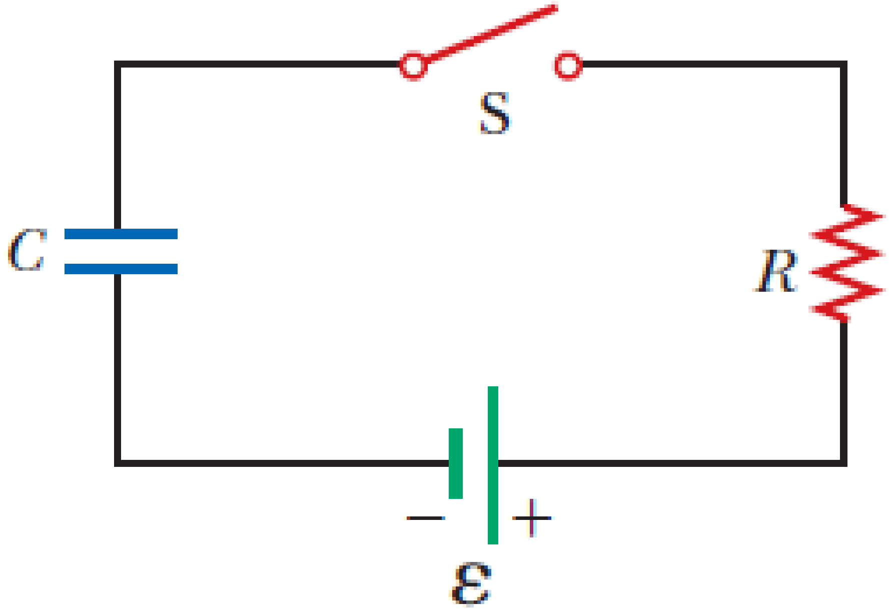


Figure P28.38

Problems 38, 67, and 68.

(a) The time constant is

$$RC = \left(1.00\times10^{6}\ \Omega\right)\left(5.00\times10^{-6}\ \text{F}\right) = \boxed{5.00\ \text{s}}$$

(b) After a long time interval, the capacitor is "charged to thirty volts," separating charges of

$$Q = C\mathcal{E} = \left(5.00\times10^{-6}\ \text{C}\right)\left(30.0\ \text{V}\right) = \boxed{150\ \mu\text{C}}$$

(c) $$I(t) = \frac{\mathcal{E}}{R}e^{-t/RC} = \left(\frac{30.0\ \text{V}}{1.00\times10^{6}\ \Omega}\right)\exp\left[\frac{-10.0\ \text{s}}{\left(1.00\times10^{6}\ \Omega\right)\left(5.00\times10^{-6}\ \text{F}\right)}\right]$$

$$= \boxed{4.06\ \mu\text{A}}$$

39. A 2.00-nF capacitor with an initial charge of 5.10 μC is discharged through a 1.30-kΩ resistor. (a) Calculate the current in the resistor 9.00 μs after the resistor is connected across the terminals of the capacitor. (b) What charge remains on the capacitor after 8.00 μs? (c) What is the maximum current in the resistor?

(a) From $I(t) = -I_0 e^{-t/RC}$,

$$I_0 = \frac{Q}{RC} = \frac{5.10 \times 10^{-6}\ \text{C}}{(1\,300\ \Omega)(2.00 \times 10^{-9}\ \text{F})} = 1.96\ \text{A}$$

$$I(t) = -(1.96\ \text{A})\exp\left[\frac{-9.00 \times 10^{-6}\ \text{s}}{(1\,300\ \Omega)(2.00 \times 10^{-9}\ \text{F})}\right] = \boxed{-61.6\ \text{mA}}$$

(b) $$q(t) = Qe^{-t/RC} = (5.10\ \mu\text{C})\exp\left[\frac{-8.00 \times 10^{-6}\ \text{s}}{(1\,300\ \Omega)(2.00 \times 10^{-9}\ \text{F})}\right]$$

$$= \boxed{0.235\ \mu\text{C}}$$

(c) The magnitude of the maximum current is $I_0 = \boxed{1.96\ \text{A}}$.

45. A charged capacitor is connected to a resistor and switch as in Figure P28.45. The circuit has a time constant of 1.50 s. Soon after the switch is closed, the charge on the capacitor is 75.0% of its initial charge. (a) Find the time interval required for the capacitor to reach this charge. (b) If $R = 250\ \text{k}\Omega$, what is the value of C?

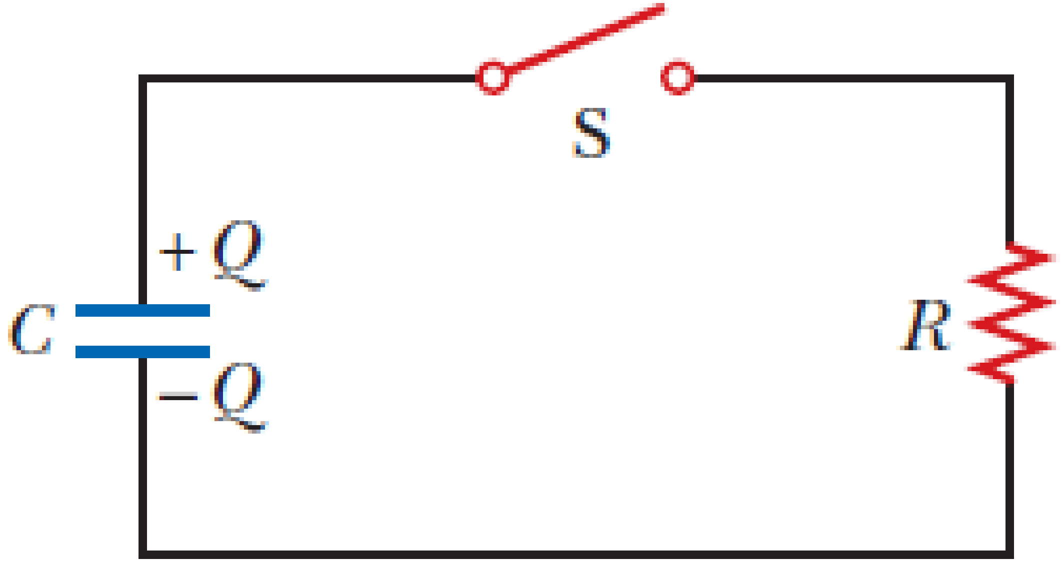


Figure P28.45

(a) The charge remaining on the capacitor after time t is $q = Qe^{-t/\tau}$.

Thus, if $q = 0.750Q$, then

$$0.750Q = Qe^{-t/\tau}$$

$$e^{-t/\tau} = 0.750$$

$$t = -\tau \ln(0.750) = -(1.50\ \text{s})\ln(0.750) = \boxed{0.432\ \text{s}}$$

(b) $\tau = RC$, so

$$C = \frac{\tau}{R} = \frac{1.50\ \text{s}}{250 \times 10^3\ \Omega} = 6.00 \times 10^{-6}\ \text{F} = \boxed{6.00\ \mu\text{F}}$$